Database – No SQL – Mongodb

Session 1 –

1. **Install MongoDB Community Edition on your system and use the mongo shell to create a new database named 'foodieApp'.**

Ans:
Install MongoDB Community Edition for your operating system from the official MongoDB documentation:

MongoDB Community Edition installation guide

After installation, start the MongoDB server (mongod) and open a terminal. Launch the shell:

```
mongosh
```

```
test>
```

2. **Inside the 'foodieApp' database, create a collection named 'restaurants' and insert a document for a Zomato-style restaurant with fields: name, cuisine, rating, and location.**

Ans:
use foodieApp :

```
db.createCollection("restaurants")
```

Insert the first restaurant:

```
db.restaurants.insertOne({
    name: "Spice Garden",
    cuisine: "Indian",
    rating: 4.5,
    location: {
        city: "Vadodara",
        area: "Alkapuri"
    }
})
```

3. **Insert three more restaurant documents into the 'restaurants' collection, making sure to use at least one field with an array data type (for example, a 'menu' array listing popular dishes).**

Ans:

```
db.restaurants.insertMany([
    {
        name: "Pizza Palace",
        cuisine: "Italian",
        rating: 4.2,
        location: {
            city: "Vadodara",
            area: "Sayajigunj"
        },
        menu: ["Margherita Pizza", "Pasta Alfredo", "Garlic Bread"]
    },
    {
        name: "Sushi House",
        cuisine: "Japanese",
        rating: 4.7,
        location: {
            city: "Ahmedabad",
            area: "Navrangpura"
        },
        menu: ["Sushi", "Ramen", "Tempura"]
    },
    {
        name: "Burger Hub",
        cuisine: "American",
        rating: 4.0,
        location: {
            city: "Vadodara",
            area: "Manjalpur"
        },
        menu: ["Classic Burger", "French Fries", "Milkshake"]
    }
])
```

4. **Retrieve all documents from the 'restaurants' collection and display them in the mongo shell. Identify which BSON types are used for each field.

Hint: Look for String, Number, Array, and Object types in your documents.**

Ans:

Run:

db.restaurants.find()

For easier-to-read output:

db.restaurants.find().pretty()

You should see documents similar to:

```
{
  _id: ObjectId('...'),
  name: 'Spice Garden',
  cuisine: 'Indian',
  rating: 4.5,
  location: {
    city: 'Vadodara',
    area: 'Alkapuri'
  }
}
{
  _id: ObjectId('...'),
  name: 'Pizza Palace',
  cuisine: 'Italian',
  rating: 4.2,
  location: {
    city: 'Vadodara',
    area: 'Sayajigunj'
  },
  menu: [ 'Margherita Pizza', 'Pasta Alfredo', 'Garlic Bread' ]
}
...
```